

KCPC Week 2 Problems

Heuristics, Advanced Topics, and Algorithmic Bridges

How To Use This Sheet

Start with the heuristic problems to complete the Pair 1 curriculum. The advanced and algorithmic sections then deepen those ideas if time permits. For algorithmic bridge problems, sketch the algorithm and argue correctness before writing code.

1 Wishful Thinking

Basic – Complete The Square

For all real numbers x and y , prove that

$$x^2 + y^2 \geq \frac{(x + y)^2}{2}.$$

Try to rewrite the difference as something that is clearly non-negative.

Easy – Make The Sum Telescope

Prove that

$$\sum_{k=1}^n \frac{1}{k(k+1)} = \frac{n}{n+1}.$$

Wish that each term were a difference of two simpler fractions.

Medium – A Number Puzzle

Construct infinitely many integer solutions to $x^2 + y^2 = 2z^2$ with $x, y, z > 0$. Wish that a Pythagorean triple were hidden in the equation.

Hard – Improve The Recurrence

Let $a_1 = 1$ and $a_{n+1} = 2a_n + 1$. Find a closed formula for a_n . Wish that the unwanted $+1$ were absorbed into the sequence.

Very Hard – Factor The Equation

For a fixed positive integer n , find all positive integer pairs (x, y) satisfying

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{n}.$$

Wish that the equation could be written as a product.

2 Symmetry

Basic – Chomp: Poisoned Chocolate

On an $m \times n$ rectangular chocolate bar, the bottom-left square is poisoned. Players alternately choose a square and eat it together with all squares above and to its right. The player forced to eat the poisoned square loses. Show that the first player has a winning strategy for any $m, n \geq 2$. Use a symmetry strategy.

Easy – Coins On A Circular Table

Two players alternately place identical non-overlapping circular coins on a circular table. The player who cannot move loses. Show that the first player has a winning strategy.

Medium – A Symmetric Integral

An integrable function $f : [0, 1] \rightarrow \mathbb{R}$ satisfies $f(x) + f(1 - x) = 1$ for every $x \in [0, 1]$. Evaluate $\int_0^1 f(x) dx$.

Hard – Odd Skew-Symmetric Matrices

Let A be an $n \times n$ real matrix satisfying $A^T = -A$. Prove that if n is odd, then $\det A = 0$.

Very Hard – Six People

Among any six people, prove that there are either three mutual acquaintances or three mutual strangers.

3 Extreme Principle

Basic – Erdős–Szekeres Monotone Subsequence

Prove that any sequence of $n^2 + 1$ distinct real numbers contains an increasing or decreasing subsequence of length $n + 1$. (Hint: associate each term with the length of the longest increasing subsequence starting there.)

Easy – Minimum Degree Forces A Cycle

Prove that every finite graph in which every vertex has degree at least 2 contains a cycle. Start with a longest path.

Medium – An Empty Triangle

A finite set of points in the plane is not contained in one line. Prove that three of the points form a triangle whose interior contains none of the other points.

Hard – A Path Through A Tournament

In a tournament, every pair of vertices is joined by exactly one directed edge. Prove that every finite tournament has a directed path that visits every vertex exactly once.

Very Hard – Longest Paths Meet

Prove that any two longest paths in a finite connected graph share at least one vertex.

4 Reduction

Basic – An Acquaintance Graph

At a party of $n \geq 2$ people, acquaintance is mutual. Prove that two people have the same number of acquaintances at the party. Reduce the problem to a graph and consider the possible vertex degrees.

Easy – The Bridges Of Königsberg

Reduce the bridges puzzle to a graph and prove that no walk crosses each of the seven bridges exactly once.

Medium – Missionaries And Cannibals

Three missionaries and three cannibals must cross a river using a boat that carries at most two people. If cannibals ever outnumber missionaries on either bank, the missionaries are eaten. Model the problem as a state graph and find a safe crossing sequence.

Hard – Lamps As Linear Algebra

n lamps are arranged in a circle. A move toggles lamp i and its two neighbours. Given an initial configuration, give an algorithm to decide whether all lamps can be turned off.

Very Hard – Assignment As Maximum Flow

There are n students and n projects. Each student supplies a list of acceptable projects. Build a flow network for assigning every student a distinct acceptable project. State how its maximum flow determines whether an assignment exists.

Advanced Material

The remaining problems are optional extensions. Their theory and worked examples are presented in *Week 2 Advanced Slides*.

5 Advanced Logic And Common Knowledge

Basic – Blue-Eyed Islanders ($k = 2$)

There are 2 islanders with blue eyes, and the rest have brown. They cannot see their own eyes. A trusted outsider announces: “At least one of you has blue eyes.” Each day, anyone who knows their own eye colour leaves at midnight. What happens and on which day?

Easy – Cheating Husbands

In a village of 100 couples, some husbands have been unfaithful. Every wife knows which *other* husbands have cheated, but not her own. The queen announces: “At least one husband has been unfaithful.” If a wife deduces her husband cheated, she must act at midnight. What happens?

Medium – Sum And Product Puzzle

Two integers x, y with $2 \leq x < y$ and $x + y \leq 100$. Mr P knows the product xy ; Mr S knows the sum $x + y$. P: “I don’t know the numbers.” S: “I knew you didn’t know.” P: “Now I know the numbers.” S: “Now I know them too.” Find x and y .

Hard – Cheryl’s Birthday

Cheryl gives Albert and Bernard a list of 10 possible dates. She whispers the month to Albert and the day to Bernard. Albert: “I don’t know the birthday, but I know Bernard doesn’t know either.” Bernard: “At first I didn’t know, but now I do.” Albert: “Now I also know.” When is Cheryl’s birthday?

Very Hard – 100 Prisoners And A Light Bulb

100 prisoners are in solitary cells. There is a central room with a light bulb (initially off) and a switch. One prisoner at a time is chosen randomly to visit the room; they may toggle the switch and/or announce “everyone has visited.” If correct, all go free; if wrong, all die. They may agree a strategy beforehand. Design a protocol that guarantees freedom.

6 Advanced Invariants And Monovariants

Basic – Peg Solitaire (Small Board)

On a 3×3 board with the centre empty, pegs jump orthogonally over adjacent pegs into empty squares, removing the jumped peg. Can you finish with one peg in the centre?

Easy – Token Game Modulo 3

Tokens on integer points. A move: choose x with tokens at x and $x + 1$; remove them and place one at $x + 2$. Start with one token at 0 and one at 1. Can you ever have a token at 100? (Hint: track $\sum 2^{\text{position}}$ modulo 3.)

Medium – Conway's Soldiers

On an infinite checkerboard, pegs occupy all squares with $y \leq 0$. A peg jumps orthogonally over another into an empty square, removing the jumped peg. Can a peg ever reach $y = 5$? (Hint: assign weights $x^{|y|}$ with $x = (\sqrt{5} - 1)/2$.)

Hard – Polynomial Invariant For $(x+2, y-1)$

A token starts at $(0, 0)$. A move changes (x, y) to $(x + 2, y - 1)$. Find a non-constant polynomial $P(x, y)$ such that $P(x + 2, y - 1) = P(x, y)$ for all x, y . Characterise all reachable positions using this invariant.

Very Hard – IMO 1986 Problem 3 (Sequence Invariant)

Given integer sequence $a_0, a_1, \dots$ with $a_0 = 0$, $a_1 = 1$, and $a_{n+1} = a_n + a_{n-1} + 1$. Prove that $a_n^2 - a_{n-1}a_{n+1}$ is invariant modulo 3, and find its value.

7 Colouring And Tiling: Generalisations

Basic – Mutilated Chessboard: Four Corners Removed

An 8×8 chessboard has all four corner squares removed. Can the remaining board be tiled by dominoes? If not, prove why not using a 3-colouring. If so, give a tiling.

Easy – Tiling With T-Tetrominoes

Can a 4×4 board be tiled by T-tetrominoes (four squares in a T shape)? Can a $4 \times n$ board for $n > 4$? Use a 4-colouring based on coordinates mod 2.

Medium – Fault-Free Domino Tiling

A domino tiling of a rectangle has a fault line if a straight line from one side to the other cuts no dominoes. Prove that a 6×6 board has a fault-free tiling, but a 6×8 board does not.

Hard – Triominoes On A Chessboard

Can an 8×8 chessboard with one corner removed be tiled by straight trominoes (1×3 rectangles)? Use a 3-colouring of the board.

Very Hard – Generalised Mutilated Chessboard

For which m, n can an $m \times n$ board with two squares of opposite colours removed be tiled by dominoes? Prove your characterisation.

8 Induction And Recursion: From Proof To Algorithm

Basic – Josephus Problem

n people stand in a circle. Eliminate every second person until one remains. Where should you stand to survive? Find a recurrence and closed form for $J(n)$.

Easy – Gray Codes

An n -bit Gray code lists all 2^n binary strings so consecutive strings differ in one bit. Give a recursive construction. Prove it works by induction.

Medium – Recursive 15-Puzzle Parity

Given a 4×4 15-puzzle configuration, design a recursive algorithm to determine if it is solvable. Use the invariant from Week 1.

Hard – Counterfeit Coin With n Weighings

Generalise the 12-coin, 3-weighing problem. With n weighings on a balance scale, what is the maximum number of coins among which you can find one counterfeit (heavy or light)? Give a strategy achieving the bound.

Very Hard – Hadamard Matrix Construction

A Hadamard matrix of order n is an $n \times n$ matrix with entries ± 1 whose rows are mutually orthogonal. Give a recursive construction for $n = 2^k$ and prove it works.

9 Algorithmic Bridges

Basic – A Maze As A Graph

You are given a rectangular grid of open and blocked cells, together with a start and finish cell. You may move one cell up, down, left, or right through open cells. Model the maze as a graph, describe how BFS finds the minimum number of moves, and explain why the answer is optimal.

Easy – Greedy Activity Selection

You have n intervals $[s_i, f_i]$. Choose a maximum-size subset of non-overlapping intervals. Prove that picking the interval with earliest finish time, then recursing, is optimal. (Exchange argument.)

Medium – DP From Induction: Coin Change

Given coin values $c_1, \dots, c_k$ and target n , find the minimum number of coins to make exactly n . Define $f(n)$ recursively, identify overlapping subproblems, and give a tabulation algorithm.

Hard – Sliding Puzzle State-Space Search

2×3 sliding puzzle with tiles 1–5 and one empty. From a given start state, find the minimum moves to reach the goal. Describe the state encoding, transitions, and why BFS works. (You may assume at most 181,440 reachable states.)

Very Hard – Greedy Scheduling With Deadlines

n jobs, each takes 1 unit of time, has a deadline d_i and profit p_i . You may schedule at most one job per time slot. A job only earns profit if completed by its deadline. Design a greedy algorithm to maximise total profit and prove it correct. (Hint: consider jobs in decreasing profit order; use a disjoint-set union to find latest free slot.)

